

Precise Modulation of Intramolecular Aggregation-induced Electrochemiluminescence by Tetraphenylethylene-based Supramolecular Architectures

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Abstract: The precisely modulated synthesis of programmable light-emitting materials remains a challenge. To address this challenge, we construct four tetraphenylethylene-based supramolecular architectures (SA, SB, SC, and SD), revealing that they exhibit higher electrochemiluminescence (ECL) intensities and efficiencies than the tetraphenylethylene monomer and can be classified as highly efficient and precisely modulated intramolecular aggregation-induced electrochemiluminescence (PI-AIECL) systems. The best-performing system (SD) shows a high ECL cathodic efficiency exceeding that of the benchmark tris(2,2'-bipyridyl)ruthenium(II) chloride in aqueous solution by nearly six-fold. The electrochemical characterization of these architectures in an organic solvent provides deeper mechanistic insights, revealing that SD features the lowest electrochemical band gap. Density functional theory calculations indicate that the band gap of the guest ligand in the SD structure is the smallest and most closely matched to that of the host scaffold. Finally, the SD system is used to realize ECL-based cysteine detection (detection limit = 14.4 nM) in real samples. Thus, this study not only provides a precisely modulated supramolecular strategy allowing chromophores to be controllably regulated on a molecular scale, but also inspires the programmable synthesis of high-performance aggregation-induced electrochemiluminescence emitters.

Introduction

Molecular luminescence and electrochemiluminescence (ECL) properties are strongly influenced by the spatial arrangement of chromophores, e.g., by the supramolecular stacking pattern and the way multiple monomers assemble into crystal structures.^[1] ECL, i.e., the emission of light due to electrochemical reactions between substances electro-generated at electrode surfaces,^[2] offers the advantages of operational and miniaturization simplicity, absence of excitation light sources, and precise control of reaction kinetics, thus finding applications in clinical diagnostics,^[3] environmental monitoring,^[4] food security,^[5] and immunoassays,^[6] and other areas requiring ultrasensitive sensing.

The commercialization of ECL-based sensing requires the development of emitters with high luminescence efficiencies and stabilities. In particular, structures favoring aggregation-induced emission (AIE) can substantially increase luminescence intensity, mainly by restricting intramolecular rotation and suppressing nonradiative deactivation pathways.^[7] Among such structures, tetraphenylethylene (TPE) features a propeller-like skeleton and exhibits the advantages of remarkable AIE performance, structural modification flexibility, and easy synthesis. TPE derivatives are widely used in ECL-based biosensing, single-molecule imaging, and other applications,^[8] whereas monodisperse TPE monomers suffer from low luminescence efficiency and are therefore poorly suited for homogeneous-phase analyses.^[9] Therefore, precisely controllable and well-dispersed molecular-scale TPE oligomers (e.g., supramolecular architectures with few-layer TPE stacking) are highly sought after. The related synthetic strategies mainly involve the formation of higher-order functional assemblies from discrete individual components after spontaneous binding through intermolecular interactions, which enables the formation of supramolecular structures with precise size/shape and predictable availability and optical properties.^[10]

Inspired by the ability of host-guest interactions to stabilize supramolecular architectures and reduce the energy gap for electron transfer,^[11] we combined four guest ligands (LA, LB, LC, and LD) with Zn²⁺ ions and a TPE-based host scaffold (HS) to form multilayered TPE stacking architectures. In these architectures, the TPE moieties could be precisely modulated using spatial arrangements, as

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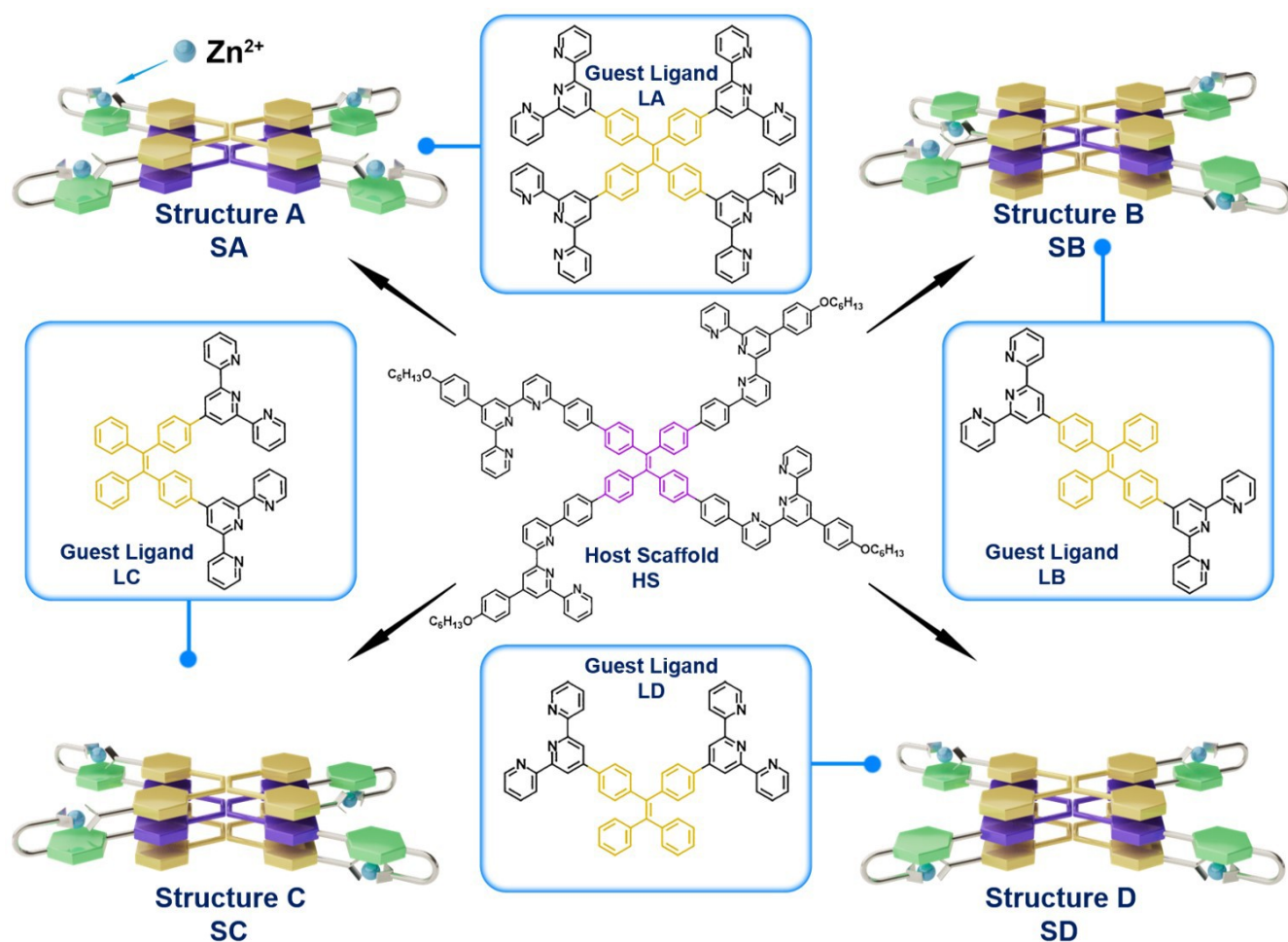
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structures with twisted noncoplanar conformations could prevent the reduction of ECL induced by intermolecular interactions due to π - π stacking.^[12] The effects of the TPE architectures on PI-AIECL properties in amphiphilic homogeneous solutions were investigated in more detail to confirm the related structure-property relationships. The high performance of rationally designed supramolecular architectures was ascribed to the substantial increase in ECL efficiency due to the modulation of emissive architectures at a slight angle. In other words, the strong ECL intensity of the TPE architectures originated not only from restricted benzene ring rotation and the resulting suppression of nonradiative relaxation, but also from the controlled limited stacking of architectures allowing the TPE monomer to be triggered as effectively as possible.^[13] In this regard, a design strategy for supramolecular architectures could provide new ideas for the development of advanced AIECL emitters.

Results and Discussion

As shown in Scheme 1, a claw-like TPE-based HS ligand and tetrapotic (LA) and ditopic (LB-LD) guest ligands were used to construct multilayer emissive architectures (SA-SD). The TPE core of the HS restricted intramolecular rotation. SB, SC, and SD differed in the rotation angle of their guest ligand (180° , 60° , and 120° , respectively) and thus exhibited different luminescence properties. Regarding the entire self-assembly process,^[14] the terpyridine (tpy) moiety exhibited a slight electron-withdrawing effect relative to the TPE core. Complexation with Zn^{2+} enhanced this effect and thus facilitated intramolecular charge transfer (ICT) from the TPE core to tpy- Zn^{2+} moieties.^[15]

As expected, the obtained supramolecular architectures exhibited PI-AIECL emission in aqueous solution. In the presence of the coreactant ($\text{K}_2\text{S}_2\text{O}_8$), the TPE assembly was modified on the electrode surface to provide an intense ECL signal (Figure 1A). The TPE assembly⁻ radical anion interacted with the sulfate radical anion ($\text{SO}_4^{\bullet-}$) through electron transfer to produce an excited state, TPE assembly*. After precisely controlled stacking, the free rotation of



Scheme 1. Assembly of multilayer tetraphenylethylene (TPE)-based emissive architectures. TPE moieties are colored to distinguish them from guest ligands.

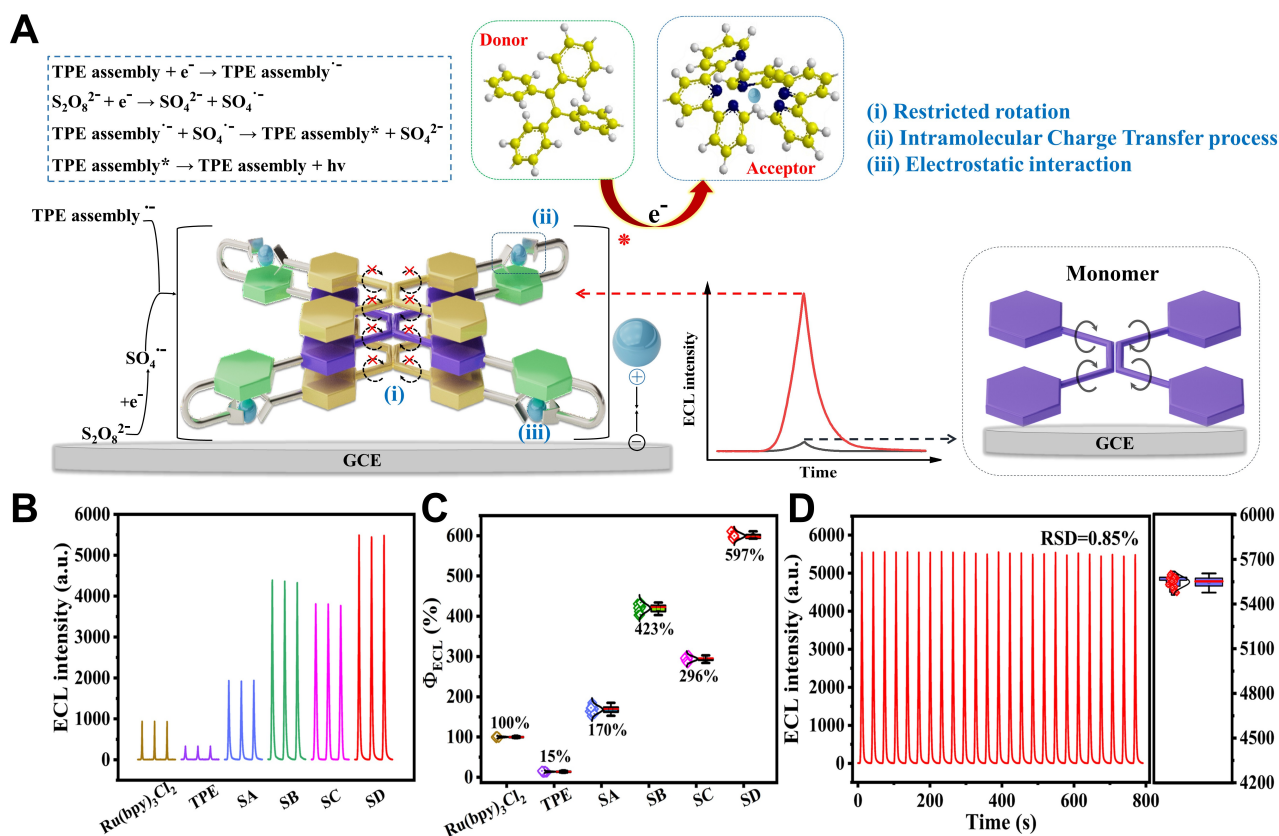


Figure 1. (A) Mechanism of electrochemiluminescence (ECL) in TPE assembly/K₂S₂O₈ systems. (B) ECL intensity–time curves and (C) ECL efficiencies obtained for 1 mM Ru(bpy)₃Cl₂, TPE, SA, SB, SC, and SD in phosphate-buffered saline (PBS) containing 0.1 M K₂S₂O₈ (values are expressed as medians). (D) (left) ECL intensity–time curve recorded for SD/K₂S₂O₈ system under optimal conditions over 25 continuous cycles and (right) the corresponding box diagram.

the supramolecular benzene ring in the molecular structure was limited. In addition, the multilayer architecture facilitated ICT from the electron donor (TPE) to the electron acceptor (tpy-Zn²⁺). As reported previously, the well-conjugated large tpy groups can restrain intramolecular rotations even in solution, thus exhibiting remarkable emission in solution and aggregated states.^[16] The formation of intramolecular donor-acceptor pairs increased the luminescence intensity in the aggregated state, which is a unique characteristic of the TPE supramolecular architecture. Therefore, the efficient ECL of the TPE assembly was ascribed to the stabilized TPE assembly^{-•} radical anion and ICT of the conjugated donor-acceptor architecture.^[2b] The luminescence intensities of the four supramolecular architectures were systematically investigated to understand their cathodic ECL efficiencies. In the presence of K₂S₂O₈, the ECL intensities of SA–SD significantly exceeded that of the TPE monomer (Figure 1B), particularly in the case of SD, which indicated the superior ECL performance of the latter architecture. To further quantify ECL performance, we calculated the ECL efficiencies (Φ_{ECL}) of the four architectures as follows:^[17]

$$\Phi_{\text{ECL}} = \left[\left(\int I_{\text{ECL}} dt / \int i dt \right)_{\text{x}} / \left(\int I_{\text{ECL}} dt / \int i dt \right)_{\text{st}} \right] \times 100\%. \quad (1)$$

where I_{ECL} and i are ECL intensity and electrochemical current, respectively, “st” refers to the Ru(bpy)₃Cl₂/K₂S₂O₈ standard (bpy=2,2'-bipyridyl) with an assumed Φ_{ECL} of 100% and an ECL intensity shown in Figure S1, and “x” refers to TPE supramolecular architectures. Under the same conditions, SD exhibited the highest ECL efficiency of ≈597% (Figure 1C). The difference in ECL efficiency originated from that in the spatial arrangement of the structures. As shown in Figure 1D, the ECL intensity of the SD/K₂S₂O₈ system remained constant during 25 continuous cycles in a phosphate-buffered saline (PBS) solution under optimal conditions (relative standard deviation (RSD)=0.85%), which proved the stability of the SD architecture. This behavior was ascribed to the electrostatic interaction of Zn²⁺ with the negatively charged electrode surface. In addition, the scanning electron microscopy (SEM) image of the modified assembly on the electrode surface and the stabilities of the other three TPE-based systems are presented in Figures S2–S3. Given that the SD system featured the strongest and most stable ECL signal, it was selected for further investigation.

Subsequently, we examined the reactivities of the electrogenerated substances obtained in the ECL process. Figure S4 shows the differential pulse voltammetry curves of 1 mM SA, SB, SC, and SD solutions in N₂-saturated PBS

containing 0.1 M KCl, indicating that the experiments in this study proceeded under the conditions of cathode potential scanning.

To further demonstrate the effect of structural differences on electrochemical properties and determine the oxidation and reduction potentials required for ECL, we probed TPE, SA, SB, SC, and SD by cyclic voltammetry (CV). Measurements were performed in a 0.1 M solution of tetrabutylammonium hexafluorophosphate (TPAPF₆) in acetonitrile purged with argon for 30 min and potentials were referenced to ferrocene/ferrocenium (Fc/Fc⁺) (detail are shown in Figure S5). The highest occupied molecular orbital (HOMO) energies (E_{HOMO}), lowest unoccupied molecular orbital (LUMO) energies (E_{LUMO}), and band gaps (E_{gap}) were calculated using Equations 2–4:^[18]

$$E_{\text{HOMO}} (\text{eV}) = -(E_{\text{onset, ox vs. Fc}^+/\text{Fc}} + 4.8), \quad (2)$$

$$E_{\text{LUMO}} (\text{eV}) = -(E_{\text{onset, red vs. Fc}^+/\text{Fc}} + 4.8), \quad (3)$$

$$E_{\text{gap}} = E_{\text{LUMO}} - E_{\text{HOMO}}. \quad (4)$$

Here, $E_{\text{onset, ox}}$ represents the onset potential of oxidation, and $E_{\text{onset, red}}$ represents the onset potential of reduction. The redox potentials and band gaps of the four supramolecular architectures are presented in Table 1 and Figure 2F. E_{gap} , which is closely related to the energy difference between excited and ground states, was smallest (2.516 eV) for SD, leading to highly efficient electron–hole recombination, facilitating the approach of a given ECL route by excited states, and favoring electron transfer and radical ion formation.^[19] Thus, the SD architecture exhibited superior ECL performance.

Subsequently, we attempted to calculate the orbital energies of TPE-based supramolecular architectures using density functional theory (DFT) and thus further investigate the effect of electronic structure on ECL emission; however, these calculations proved to be very difficult. Consequently, we constructed the molecular orbital plots of the HS and

Table 1: Redox potentials and band gaps of TPE, SA, SB, SC, and SD.

Compound	$E_{\text{onset, ox vs. Fc/Fc}^+}$ [V]	$E_{\text{onset, red vs. Fc/Fc}^+}$ [V]	E_{HOMO} [eV]	E_{LUMO} [eV]	E_{gap} [eV]
TPE	1.848	−1.824	−6.648	−2.976	3.672
SA	1.891	−1.688	−6.691	−3.112	3.579
SB	1.924	−0.718	−6.724	−4.082	2.642
SC	2.187	−0.494	−6.987	−4.306	2.681
SD	1.709	−0.807	−6.509	−3.993	2.516

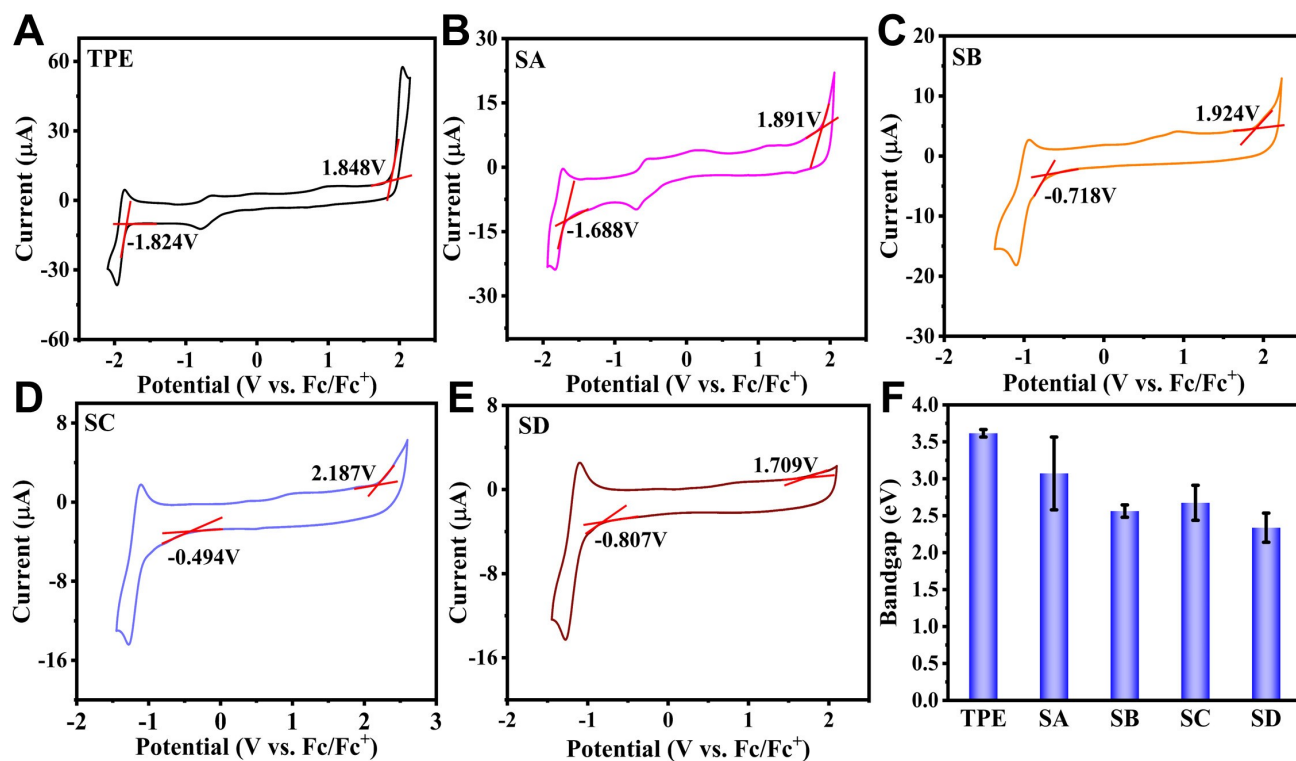


Figure 2. Cyclic voltammograms of 1 mM (A) TPE, (B) SA, (C) SB, (D) SC, and (E) SD recorded in acetonitrile/benzene containing 0.1 M tetrabutylammonium hexafluorophosphate. (F) Electrochemical band gaps of TPE-based supramolecular architectures extracted from (A–E).

guest ligands (Figure 3). The electronic structure of some ligands can partially affect molecular orbital energy levels, i.e., the HOMO and LUMO gap and thus emissions of the TPE supramolecular architectures.^[20] Substituents with extensive conjugation (e.g., tpy) favor the formation of electrochemically stable radicals as long as more conjugated groups are introduced at the TPE center.^[21] The results indicated that the HOMO and LUMO were delocalized over the TPE moiety and featured a substantial contribution from the chromophore. Thus, the TPE group had a stronger effect on structural properties than the tpy group. The HOMO–LUMO transition between TPE and tpy corresponds to intramolecular charge transfer.^[22] The smaller HOMO–LUMO gap of LD compared with those of LA, LB, and LC led to high electron affinity. The energy level of HS (LUMO = −2.65 eV, HOMO = −4.46 eV) matched that of LD (LUMO = −2.81 eV, HOMO = −4.45 eV), which facilitated electron transfer to produce the strongest ECL signal. Thus, a small HOMO–LUMO gap resulted in increased ECL intensity, which provides guidance for constructing more efficient TPE-based supramolecular architectures.^[8d]

Further insights into the mechanism of PI-AIECL enhancement were obtained by subjecting solutions of TPE, SA, SB, SC, and SD in acetonitrile to CV and ECL measurements. Figures S6–S10 A show the CV curves of 1 mM TPE and related supramolecular architectures in CH₃CN/benzene solution (v/v = 1:1) containing 0.1 M TBAPF₆ as a supporting electrolyte. When the potential was scanned in the negative direction, each architecture exhibited one reversible redox process, i.e., the generated radical anions had good stability.^[18b] ECL was simultaneously measured to explore the annihilation system, which directly reflects the inherent redox characteristics of the emitters.

The annihilation reaction produces ground and excited states, and the excited species emit light and return to the ground state. Hence, in ECL annihilation systems, the emitter is the only source of excited states.^[23] In fact, the ECL signals of TPE and its supramolecular architectures are weak because of the free rotation of the AIE-active architectures in the solution state, causing a certain non-radiative energy loss.^[24]

In view of the above, the introduction of a coreactant into the solution appears logical. Unidirectional potential scanning reduces the time delay of radical ions encountered to produce free radicals, enhancing ECL emission.^[25] Hence, we attempted to enhance the ECL by introducing a coreactant, benzoyl peroxide (BPO), into the annihilation systems (Figures S6–S10B). BPO increased the ECL intensity of all architectures (Figure S11), with the highest intensity observed for SD. Ultimately, the annihilation state is in sharp contrast with the coreaction state. Simultaneously, we recorded the CV curves of SA, SB, SC, and SD at different scan directions and rates (Figures S12–S15). The above research detailed the PI-AIECL characteristics of TPE-based supramolecular architectures and showed why the SD architecture had the highest ECL efficiency.

The photophysical properties of SA, SB, SC, and SD were investigated using ultraviolet-visible (UV-vis) absorption, fluorescence (FL), and ECL spectroscopy. The UV/Vis absorption spectra (Figure 4A) featured two peaks at 280 and 325 nm corresponding to the π – π^* transitions of the tpy group and TPE core.^[26] The SA architecture produced weak emissions. When the molar absorption coefficient was constant, SA had the smallest molecular weight, and its absorption strength was significantly lower than those of the other architectures. Notably, the absorption spectra exhib-

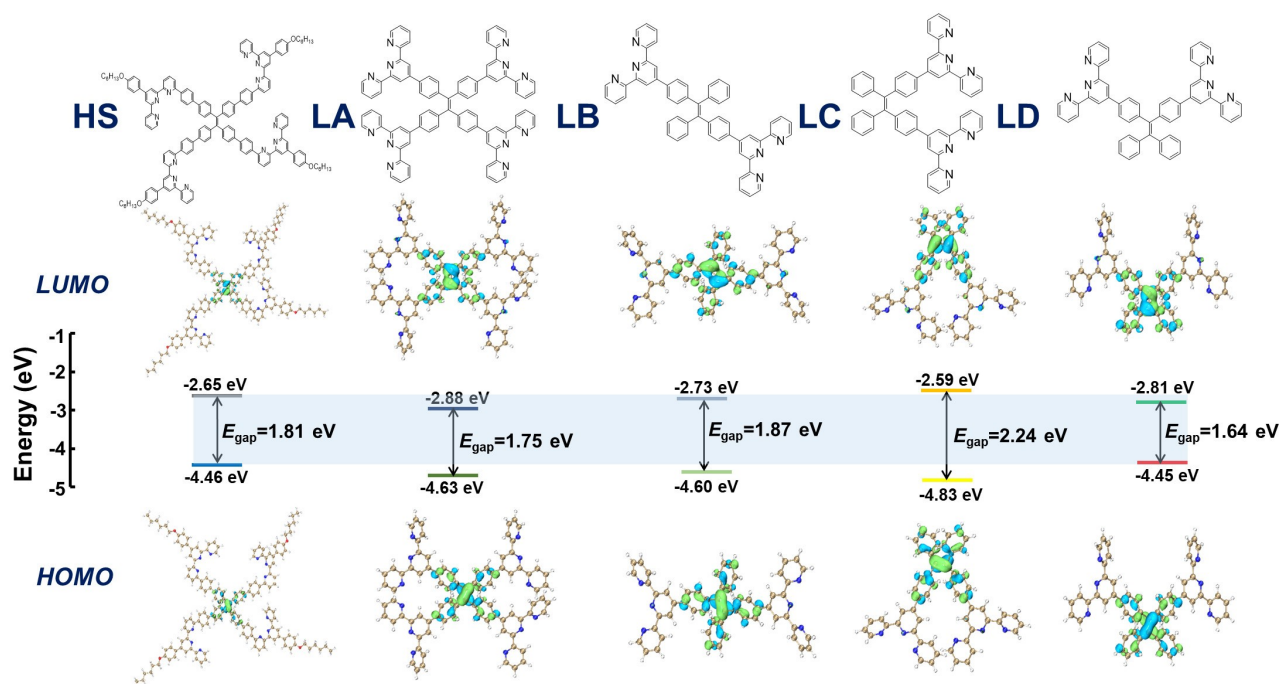


Figure 3. Amplitude plots and energies of frontier molecular orbitals calculated for different ligands using density functional theory.

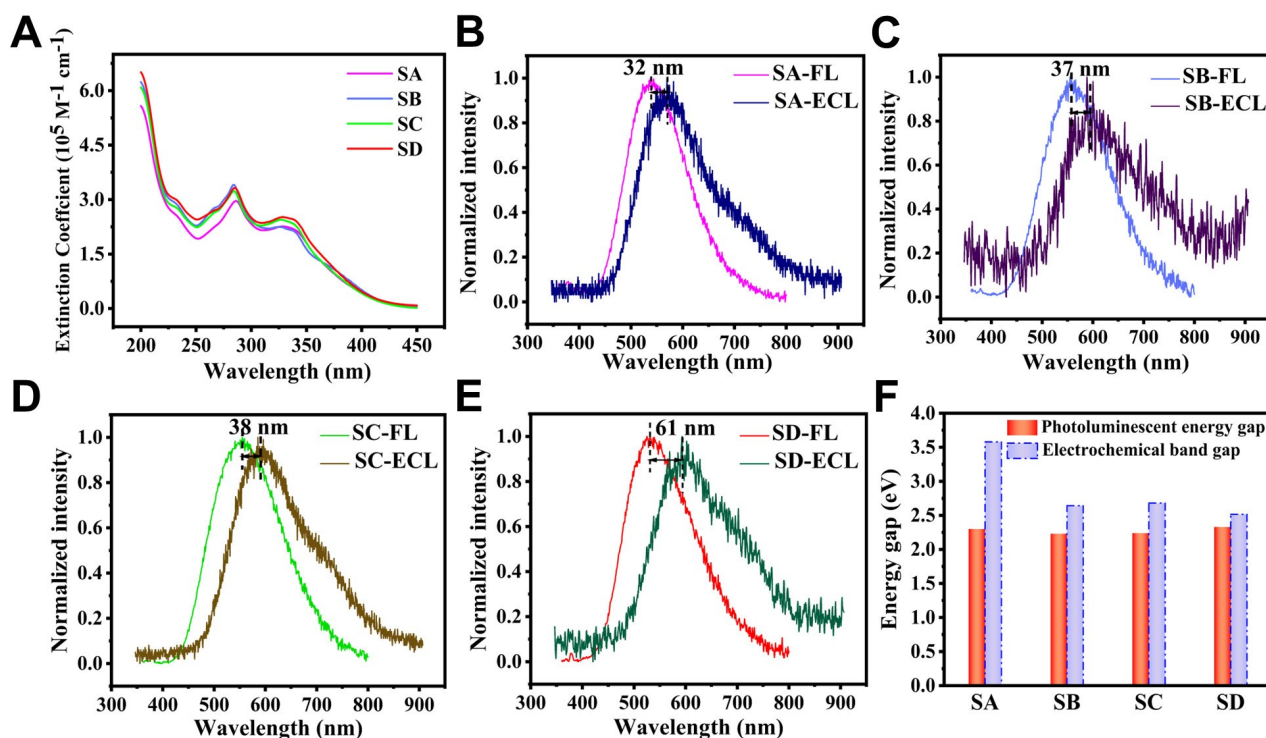


Figure 4. (A) Ultraviolet-visible absorption spectra of 1 mM SA, SB, SC, SD in acetonitrile ($\lambda_{\text{ex}} = 340$ nm). (B–E) Normalized fluorescence (FL) and ECL spectra of SA, SB, SC, and SD in solution state. (F) Comparison of photoluminescent and electrochemical band gaps of TPE-based supramolecular architectures.

ited almost indistinguishable peaks, which suggested that precise modulation did not affect the original properties of TPE architecture.

Subsequently, we compared the normalized FL and ECL spectra of SA, SB, SC, and SD, with the corresponding ECL emission maxima observed at 570, 594, 590, and 592 nm, respectively. Compared to the FL spectra, the ECL spectra featured broader red-shifted peaks (Figures 4B–E), which was ascribed to excimer emission. Excimer emission refers to the phenomenon in which excimer molecules release light as they transition from an excited state to a ground state. The emission spectrum of excimer emission is relatively broad, thus resulting in wider red-shifted peaks in ECL spectra.^[27] The four architectures with different stacking patterns exhibited similar ECL emissions, and their substantially red-shifted ECL peaks indicated that the excited states in different stacking patterns might be governed by the mechanism of surface state transitions with lower energy rather than the band gap.^[19b,28] In addition, the results were influenced by analyte concentration as well as instrument slit width and resolution.^[29] The energy gaps between the excited and ground states (E_s) were calculated from the FL peak maxima (λ) using Equation 5 (Table 2, Figure 4F):^[2d,30]

$$E_s \text{ (eV)} = 1239.81/\lambda \text{ (nm)}. \quad (5)$$

The fact that electrochemical band gaps exceeded FL emission energies implied the occurrence of nonradiative relaxation from electron promotion to the excited state.^[25]

Table 2: Photoluminescence spectroscopic data and energy gaps (E_s).

Compound	λ [nm]	E_s [eV]
SA	538	2.30
SB	557	2.23
SC	552	2.24
SD	531	2.33

Figure S16 compares the FL spectra of the HS and guest ligands, while Figure S17 presents the simulated the FL spectra of the four architectures. In addition, the spectrums of quantum efficiency and lifetime in solid state are shown in Figures S18 and S19.

The introduction of coreaction accelerators can effectively improve ECL efficiency and promote the practical application of ECL-based sensing in biological systems, as exemplified by cobalt-iron layered double hydroxide^[31] and Fe@Fe₂O₃ nanowires.^[32] In this study, MnO₂ nanoflowers (NFs) were introduced as a coreaction accelerator to enhance the ECL reaction kinetics and emission intensity (Scheme S1A). Scheme S1B was used to establish the ECL enhancement mechanism. The successful preparation of MnO₂ NFs was confirmed by SEM and X-ray diffraction analyses (Figures S20A, B). The results of ECL intensity measurements (Figure S21) indicated that MnO₂ NFs resulted in ECL properties superior to those observed for the bare electrode.

To explain the ECL enhancement by MnO₂ NFs, we recorded the CV and ECL-potential profiles of different

modified electrodes in the potential range from -1.6 to 0 V (Figures S22A–E). The results demonstrated that MnO_2 NFs could interact with $\text{S}_2\text{O}_8^{2-}$ to produce $\text{SO}_4^{\bullet-}$ (Figure S22F) and implied that MnO_2 NFs acted not only as a coreaction accelerator to promote the reduction of $\text{S}_2\text{O}_8^{2-}$, but also as a substrate to capture large amounts of TPE-based supramolecular architectures, thereby increasing ECL intensity.^[33] Thus, the coreaction accelerator indeed improved the ECL performance of the emitter, allowing it to exhibit extremely strong ECL signals.

Given the excellent and stable AIECL performance of the SD architecture, we selected three biothiols, namely glutathione (GSH), cysteine (Cys), and homocysteine (Hcy), as detection objects playing important roles in disease diagnosis and drug analysis.^[34] Figure 5A shows the related detection results, revealing that the above analytes induced a marked decrease in the ECL intensity of the functionalized SD system and could therefore be sensitively detected by the same. And other amino acids also elicited responses, which, however, were weaker than those elicited by the three biothiols. Principal component analysis (PCA) demonstrates the excellent discrimination ability for three biothiols. The ECL intensity profile of the modified GCE is shown in Figure S23, with the RSD equaling 2.65 %. The solution-phase $\text{S}_2\text{O}_8^{2-}$ was reduced to $\text{SO}_4^{\bullet-}$, which, in turn, engaged in electron transfer with biothiols to induce ECL quenching.^[35] In this study, we focused on the detection of Cys. The ECL intensity decrease of the functionalized GCE (ΔI) linearly increased with the logarithm of Cys concentration ($\lg C$) as $\Delta I = 1640.2 \lg C + 3898.3$ ($R^2 = 0.9981$) (Figures 5B, C).^[36] The limit of detection (LOD) was calculated as 14.4 nM (signal-to-noise ratio = 3), with calculation details provided in the Supporting Information. The Cys

detection performances are provided in Table S1, indicating the reasonable sensitivity of the functionalized GCE. The practical applicability of the proposed assay was examined by the quantitation of Cys in human serum using the standard addition method. The serum samples were diluted 500-fold with PBS and spiked with Cys at different concentrations. As shown in Figure 5D and Table S2, moderate to good recoveries were achieved (95.4–115.0 %). These results verify the feasibility and serviceability of the proposed assay for the quantitation of Cys in biological samples.

Conclusion

TPE-based supramolecular architectures were shown to exhibit highly efficient PI-AIECL, which was attributed to the following three factors: (i) the free rotation of the supramolecular benzene ring was restricted to minimize nonradiative transitions, (ii) ICT within the conjugated donor-acceptor architecture prolonged the lifetime of the excited state, and (iii) electrostatic interactions between Zn^{2+} ions and the electrode surface improved the stability of the emitter on the electrode surface under cathode scanning conditions. The electrochemical band gap and DFT calculations revealed that the ECL performance of SD was superior to that of the other three structures. This enhanced performance was ascribed to the lower electrochemical gap and smaller HOMO–LUMO energy gap of SD and the resulting ease of energy transfer. The AIECL performances of the different architectures obtained using a precisely modulated supramolecular strategy were substantially superior to that of the TPE monomer.

In addition, we examined the role of a coreaction accelerator to provide guidance for the appropriate introduction of such accelerators under the condition of the limited ECL intensity of the emitter and realized the ultrasensitive detection of Cys (LOD = 14.4 nM) in human serum as a proof-of-concept. Thus, this work not only demonstrates an excellent emitter for AIECL systems, but also significantly contributes to the development of a detection system for Cys, which has diverse physiological functions in living organisms.

Experimental Section

Experimental details are provided in the Supporting Information.

Acknowledgements

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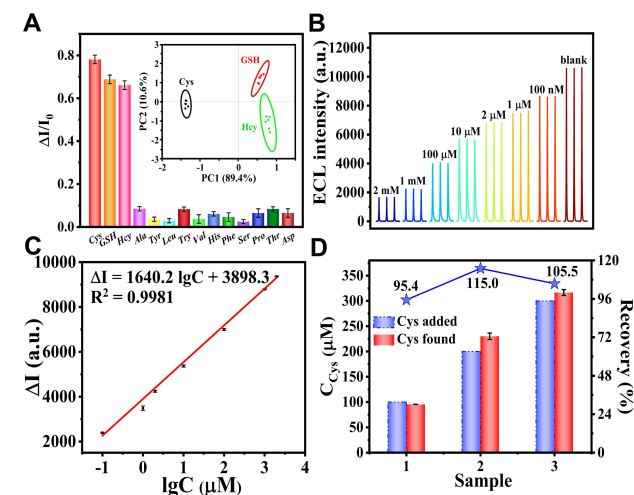


Figure 5. (A) ECL responses of modified electrodes to cysteine (Cys), glutathione (GSH), and homocysteine (Hcy), and selected amino acids. Inset: Principal component analysis (PCA) of three biothiols (TPE and SD architecture are selected). (B) ECL responses of functionalized electrodes to different concentrations of Cys (2 mM, 1 mM, 100 μM , 10 μM , 2 μM , 100 nM, blank). (C) Calibration plot of ΔECL intensity vs. logarithm of Cys concentration. (D) Detection of Cys in real samples.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Electrochemiluminescence • Intramolecular AIE • Supramolecular Strategy • Tetraphenylethylene

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